



Robust statistical recognition and reconstruction scheme based on hierarchical Bayesian learning of HRR radar target signal[☆]



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ABSTRACT

A hierarchical Bayesian model is developed to characterize the complex-valued high range resolution (HRR) radar target signal, motivated by the problem of radar automatic target recognition (RATR) robust to low signal-to-noise ratio (SNR) or narrowband interference. Here we assume a sparseness-promoting prior on the complex echoes and a Markov dependency for the location of the dominant scattering center between consecutive HRR signals. The number of the dominant scattering centers can be automatically determined, while the posterior distributions of their complex coefficients and locations can be inferred via such Bayesian model. In the training stage, based on the proposed Bayesian model, the statistical aspect-frame template can be learned for HRR complex training samples from each target-aspect sector under high SNR and without any interference. Considering the low SNR or narrowband interference problem for a test sample, the aspect-frame templates can be updated to make a robust recognition decision for the noised and interfered test sample. Simultaneously, the test sample can be denoised and recovered via the analytically posterior estimation for the reconstruction, which is referred to as the statistical compressive sensing (CS) inversion. In contrast to the traditional CS methods that only utilize the underlying sparse property of a measurement (here a measurement is a test sample), our statistical CS inversion also exploits the statistical information of the training samples, which are under high SNR and without any interference. Therefore, the better recognition and reconstruction performances can be obtained via our method. Efficient inference is performed via variational Bayesian (VB) for the proposed Bayesian model. To validate the formulation, we present our experimental results on the measured HRR dataset, with comparisons to some state-of-the-art methods.

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1. Introduction

Since the high range resolution (HRR) radar target signals contain abundant target information, such as target size, scatterer distribution, etc., HRR radar target recognition has received intensive attention from the radar automatic target recognition (RATR) community (Avci, 2013; Bharadwaj, Runkle, Carin, Berrie, & Hughes, 2001; Chen, Liu, & Bao, 2008; Chen, Liu, Chai, & Bao, 2009; Du, Liu, Bao, & Xing, 2005; Du, Liu, Bao, & Zhang, 2006; Du, Liu, & Bao, 2008b; Du et al., 2011; Du et al., 2012; Fasoula & van Genderen, 2011; Ferrara, Arnold, & Stuff, 2009; Jacobs, 1999; Nelson, Starzyk, & Ensley, 2002; Shi, Wang, Liu, Xu, & Bao, 2011;

Shang, Jiao, & Wang, 2012; Williams, Westerkamp, Gross, & Palomino, 2000; Webb, 2000; Zhang, Nasrabadi, Zhang, & Huang, 2012). One approach for HRR target recognition is to directly utilize the real profiles of the one-dimensional (1-D) HRR target signals (Du et al., 2006, 2008b, 2011; Jacobs, 1999; Shi et al., 2011; Webb, 2000), which are commonly referred to as high-resolution range profiles (HRRPs), or to extract feature vectors from the 1-D HRR target signals (Avci, 2013; Bharadwaj et al., 2001; Chen et al., 2008, 2009; Du et al., 2005, 2012; Ferrara et al., 2009; Nelson et al., 2002). Another approach is to form a synthetic radar (SAR) or inverse SAR (ISAR) image based on a sequence of HRR target signals with subsequent classification performed in the image domain (Fasoula & van Genderen, 2011; Zhang et al., 2012), of which the SAR image is usually used for the stationary target and the ISAR image for the moving target. The main advantages of 1-D HRR target recognition over SAR/ISAR image recognition are: (a) it is more efficient to make the recognition decision with a single HRR sample in the classification stage, without explicitly forming an image; (b) 1-D HRR target recognition can also be

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applied to some multi-mode radars without imaging function, e.g., the mechanical scanning radar. In the following experiments, we mainly discuss the recognition problem of the flying airplane targets. Due to the above reasons, the signal model and recognition scheme in this paper focus on 1-D HRR signal, which may also be modified for SAR/ISAR image recognition problems.

Several literatures (Bharadwaj et al., 2001; Du et al., 2006, 2008b, 2011; Jacobs, 1999; Shi et al., 2011; Webb, 2000) showed that statistical recognition is an efficient method for HRR real data (i.e., HRRP) based RATR. By statistical recognition is meant that the class membership of a measurement HRRP sample $\mathbf{x}$ is determined based on the posterior probabilities of all class memberships $p(d | \mathbf{x})$ ($d \in \{1, 2, \dots, D\}$ denotes class membership). According to the Bayes' theorem, we have $p(d | \mathbf{x}) \propto p(\mathbf{x} | d)p(d)$, where $p(\mathbf{x} | d)$ is the class-conditional likelihood and $p(d)$ is the prior class probability. Since the prior class probability is usually assumed to be uniformly distributed, estimation of the posterior probability turns into estimation of the class-conditional likelihood. Some parametric statistical models are developed to describe the class-conditional likelihood of HRRP data, e.g., Gaussian model (Jacobs, 1999), Gamma model (Webb, 2000), the two-distribution compounded model comprising Gamma distribution and Gaussian mixture distribution (Du et al., 2006), factor analysis (FA) model (Du et al., 2008b), local factor analysis (LFA) model (Shi et al., 2011), and the multitask learning (MTL) based hidden Markov model (HMM) (Du et al., 2011). In addition, HMM is also utilized to characterize the feature vector extracted from the HRRP sequence in Bharadwaj et al. (2001).

For the RATR problem, the training data are usually collected under the condition of high signal-to-noise ratio (SNR) and without any interference via some cooperative measurement experiments or directly via simulations; while the test samples are usually got in the non-cooperative circumstance (e.g., at the battle time), where the high SNR and no interference condition cannot be guaranteed due to the long distance away from the non-cooperative targets and some interference signals from the measurement system or environment (e.g., some communication signals with the frequencies within our radar band). Therefore, in order to get the robust recognition performance via a statistical recognition method, the statistical model learned in the training stage, $p(\mathbf{x} | d)$, should be modified to match the variation of the measurement conditions (including SNR and interference) in the classification stage. For the radar signal processing, the noise or interference signal directly takes effects on the complex-valued target echo, which can be expressed as a linear model, i.e., $\bar{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{v}$ with $\bar{\mathbf{x}} \in \mathbb{C}^{L \times 1}$ denoting the complex-valued noised signal without the interfered components, $\mathbf{A} \in \{0, 1\}^{L_1 \times L}$ denoting a binary projection matrix induced by the interference signals ($L_1 < L$ and $\mathbf{A}\mathbf{A}^T = \mathbf{I}_{L_1}$), $\mathbf{x} \in \mathbb{C}^{L \times 1}$ denoting the complex-valued signal under high SNR and without any interference, and $\mathbf{v} \in \mathbb{C}^{L_1 \times 1}$ denoting the complex-valued noise variable introduced from the test condition. Based on the learned $p(\mathbf{x} | d)$ and such a linear structure, the statistical model $p(\bar{\mathbf{x}} | d)$ can be easily and accurately derived in the complex domain. However, the absolute value operator to obtain the real-valued feature completely changes the linear structure, i.e., $|\bar{\mathbf{x}}| = |\mathbf{A}\mathbf{x} + \mathbf{v}| \neq \mathbf{A}|\mathbf{x}| + |\mathbf{v}|$. Thus it is difficult to accurately get the update statistical model in the real-valued domain, i.e., calculating $p(|\bar{\mathbf{x}}| | d)$ from $p(|\mathbf{x}| | d)$, to match the variation of SNR or interference condition. Based on the additive real-valued noise assumption, Du et al. (2012) proposes a noise-robust modification method for MTL-FA model. Nevertheless, according to the results in Du et al. (2012), although the noise robustness of the modified MTL-FA model is superior to those of the conventional statistical models, the proposed algorithm still cannot yield

the good recognition performance when SNR < 15 dB, since the noise model assumption is not reasonable enough.

In the RATR community, there has been some research work on target recognition based on HRR complex data (Du, Liu, & Bao, 2008a; Jouny, Garber, & Moses, 1995; Totsky, Kurbatov, Lukin, Egiazarian, & Astola, 2003; Wang, Dai, Pan, Du, & Liu, 2011; Wong, 2004). Since the bispectrum feature of complex radar signal can suppress additive disturbances that are described by symmetric probability density functions, e.g., additive white Gaussian noise, additive colored Gaussian noise, and additive non-Gaussian noise, (Jouny et al., 1995; Totsky et al., 2003) utilized such bispectrum features to improve the RATR performance under the low SNR condition. Without considering of the noise-robustness problem, (Du et al., 2008a; Wang et al., 2011; Wong, 2004) extracted some initial-phase invariant features from HRR complex data for RATR. These work (Du et al., 2008a; Wang et al., 2011; Wong, 2004) thought that the phase differences among the range cells can represent the scatterer distribution information of the corresponding target and should be useful in target discrimination. Therefore, as shown in the related work (Du et al., 2008a; Jouny et al., 1995; Totsky et al., 2003; Wang et al., 2011; Wong, 2004), there are two advantages of HRR complex data based RATR over HRR real data based RATR, i.e., noise-robustness and phase information.

The research reported here seeks a signal model for HRR complex data based on hierarchical Bayesian modeling firstly. Then a robust statistical recognition and reconstruction scheme is developed. Our contributions are as follows:

- A hierarchical Bayesian model for complex-valued HRR radar target signal is developed to infer the posterior distributions of the scattering coefficients and dominant scatterers' locations with the automatic model selection. In a Bayesian framework, the sparseness-promoting prior is imposed on the complex coefficients of the returned echoes from the scattering centers, and a Markov dependency for the location of the dominant scattering center between consecutive HRR signals is also readily embedded. The proposed signal model can automatically determine the number of dominant scattering centers (i.e., automatic model selection) via the imposed sparse prior to avoid overfitting or underfitting in the model learning.
- We present a statistical recognition scheme for HRR complex data, which is robust to the condition of low SNR or narrowband interference. Based on the proposed Bayesian signal model, the probability density function (pdf) of the HRR complex data from each target-aspect sector can be analytically estimated, which can be applied to RATR as the aspect-frame template. Such a statistical model in the complex-valued domain can be easily and accurately updated to match a test sample under low SNR or with narrowband interference, which is an obvious advantage of the statistical recognition for HRR complex data over that for HRRP data or their real-valued features.
- A statistical compressed sensing (CS) inversion is derived for reconstruction of the test sample under low SNR and with narrowband interference, which is superior to the traditional CS methods due to utilizing the statistical information learned in the training stage (beyond just sparseness). CS also known as sparse recovery, is one of the most active research areas in the recent years. One of the key contributions of CS is that stable recovery of compressible, noisy signals can be achieved through the solution of the computationally tractable sparse constrained inverse problem. Since radar imaging is typically an ill-posed linear inverse problem, the CS theory has been applied to radar imaging problems recently (Cetin & Karl, 2001; Patel, Easley, Healy, & Chellappa, 2010; Zhang et al., 2010). Nevertheless, all these papers only utilize the underlying sparse property of a set of HRR signals corresponding to an image to achieve the recovery

of the radar image. Here our statistical CS method is performed in the classification stage of the supervised recognition scheme, therefore, not only the sparse property but also the statistical information learned under high SNR and without any interference can be utilized in the reconstruction algorithm. The CS experiments based on the measured HRR dataset show the inspiring results via our method.

- Efficient inference is performed via variational Bayesian (VB) for the proposed Bayesian model.

The remainder of the paper is organized as follows. We introduce the Bayesian model for HRR complex data in Section 2. The statistical recognition scheme and CS inversion are derived in Section 3. The detailed experimental results based on measured HRR data are provided in Section 4, followed by conclusions in Section 5.

2. Bayesian signal model

2.1. Signal model construction

As discussed in Du et al. (2005, 2006, 2008b, 2011, 2012), Jacobs (1999) and Webb (2000), HRR scattering from a complex target yields target signature that is a strong function of the target-sensor orientation, referred to as the target-aspect sensitivity in Du et al. (2005, 2006, 2008b, 2011, 2012) and Shi et al. (2011). To deal with the target-aspect sensitivity, the multi-aspect HRR dataset from each target is divided into subsets (as in Du et al., 2005; Du et al., 2006; Du et al., 2008b; Du et al., 2012), where the HRR data in each subset are collected from a target-aspect sector roughly without the scatterers' motion through range cells (MTRC). Thus each subset is defined to be an aspect-frame from the target (Du et al., 2005, 2006, 2008b, 2012).

According to Keller (1962), the high-frequency scattering response of an object is well approximated as a sum of responses from individual scattering centers (Keller, 1962). Therefore, the frequency response from the n th sample in the m th aspect-frame of the target d (here $n \in \{1, 2, \dots, N\}$ with N denoting the number of samples in each aspect-frame, $m \in \{1, 2, \dots, M_d\}$ with M_d denoting the number of aspect-frames of the d th target, and $d \in \{1, 2, \dots, D\}$ with D denoting the number of targets) can be represented by

$$y_n^{(d,m)}(l) = \sum_{k=1}^K w_n^{(d,m)}(k) \cdot \exp\left(-j2\pi \frac{2r_k}{c}(l-1)\Delta f\right) + \eta_n^{(d,m)}(l) \quad (1)$$

where $y_n^{(d,m)}(l)$ represents the l th frequency response from the n th sample in the m th aspect-frame of the target d with $l \in \{1, \dots, L\}$ and L denoting the number of frequency components, $w_n^{(d,m)}(k)$ represents the corresponding scattering coefficient from the k th range cell with $k \in \{1, \dots, K\}$ and K denoting the number of range cells, c denotes the propagation velocity of radar signal, $\{r_k\}_{k=1}^K$ denote the locations of range cells, Δf denotes the frequency interval between the neighboring frequency components, and $\{\eta_n^{(d,m)}(l)\}_{l=1}^L$ represent the noise components. Let

$$\phi(r_k) = \left[1, \exp\left(-j\frac{4\pi}{c}r_k\Delta f\right), \exp\left(-j\frac{4\pi}{c}r_k2\Delta f\right), \dots, \exp\left(-j\frac{4\pi}{c}r_k(L-1)\Delta f\right)\right]^T, \Phi = [\phi(r_1), \phi(r_2), \dots, \phi(r_K)] \quad (2)$$

Then, the signal model in (1) can be expressed in a vector-matrix form as

$$\mathbf{y}_n^{(d,m)} = \Phi \mathbf{w}_n^{(d,m)} + \boldsymbol{\eta}_n^{(d,m)} \quad (3)$$

where $\mathbf{y}_n^{(d,m)} = [y_n^{(d,m)}(1), y_n^{(d,m)}(2), \dots, y_n^{(d,m)}(L)]^T$, $\boldsymbol{\eta}_n^{(d,m)} = [\eta_n^{(d,m)}(1), \eta_n^{(d,m)}(2), \dots, \eta_n^{(d,m)}(L)]^T$, $\mathbf{w}_n^{(d,m)} = [w_n^{(d,m)}(1), w_n^{(d,m)}(2), \dots, w_n^{(d,m)}(K)]^T$ denotes the n th HRR complex sample in the m th aspect-frame of the target d , and $|\mathbf{w}_n^{(d,m)}|$ is the corresponding HRRP sample. When $K = L$, Φ is a complete orthogonal basis matrix; when $K > L$, Φ is a non-orthogonal and redundant dictionary matrix, then the superresolution representation can be achieved. Below we explain the model in detail.

2.1.1. Scattering coefficients

As discussed in Zhang et al. (2010) for CS-based ISAR imaging, the target signal can be approximated by the echoes from the dominant scattering centers which are assumed to be sparsely distributed in the target, while those from the weak scattering centers are regarded as the noise components. Therefore, to seek sparseness on the scattering coefficients from each HRR sample, we assume $\mathbf{w}_n^{(d,m)} = \hat{\mathbf{w}}_n^{(d,m)} \circ \mathbf{z}_n^{(d,m)}$, where $\circ$ represents the point-wise (Hadamard) vector product and $\mathbf{z}_n^{(d,m)}$ is a binary vector. This approach separates the learning of sparseness from the learning of coefficient values, such that the zero component in $\mathbf{w}_n^{(d,m)}$ is exactly zero.

The first binary vector in an aspect-frame, e.g., $\mathbf{z}_1^{(d,m)}$, is modeled as

$$z_1^{(d,m)}(k) \sim \text{Ber}(u_{k,1}^{(d,m)}), \quad u_{k,1}^{(d,m)} \sim \text{Beta}(e_0, f_0) \quad (4)$$

where $z_1^{(d,m)}(k)$ denotes the k th component in $\mathbf{z}_1^{(d,m)}$, $\text{Ber}(\cdot)$ and $\text{Beta}(\cdot)$ refer respectively to the Bernoulli and Beta densities parameterized by the arguments inside parentheses, and hyperparameters $e_0 > 0$, $f_0 > 0$. Since the density function $\text{Beta}(e_0, f_0)$ is the conjugate prior for $u_{k,1}^{(d,m)}$, this Bernoulli-Beta prior allows the closed-form Bayesian inference for $z_1^{(d,m)}(k)$, which is a zero/one indicator variable. Note that $\mathbb{E}[u_{k,1}^{(d,m)}] = e_0/(e_0 + f_0)$ and $\mathbb{C}[u_{k,1}^{(d,m)}] = (e_0 f_0)/[(e_0 + f_0)^2(e_0 + f_0 + 1)]$, where $\mathbb{E}[\cdot]$ denotes the expectation of a random variable and $\mathbb{C}[\cdot]$ denotes the variance of a random variable. To impose sparseness on $\mathbf{z}_1^{(d,m)}$, we may let $e_0/(e_0 + f_0) \approx 0$, and therefore it is probable that $u_{k,1}^{(d,m)}$ is close to zero. Since K is usually a large integer, we specify $e_0 = 1/K$ and $f_0 = (K-1)/K$ in our experiments, same as the corresponding priors used in Ding, He, and Carin (2011), then $\mathbb{E}[u_{k,1}^{(d,m)}] \approx 0$ with a very small variance $\mathbb{C}[u_{k,1}^{(d,m)}]$.

Considering the temporal dependency between HRR samples from a given aspect-frame, if there is a dominant scattering center in a specific range cell in a sample, it is very likely that this dominant scattering center will appear in about the same range cell in the next sample. Thus we may exploit the belief that if $z_n^{(d,m)}(k)$ is non-zero, the corresponding binary component for the next sample $z_{n+1}^{(d,m)}(k)$ is also non-zero with a high probability. Here we assume the temporal relationship across $\{z_n^{(d,m)}\}_{n=2}^N$ satisfies a simple Markov structure.

$$z_n^{(d,m)}(k) \sim \text{Ber}(u_{k,n}^{(d,m)}), \quad u_{k,n}^{(d,m)} \sim \begin{cases} \text{Beta}(e_0^0, f_0^0) & \text{if } z_{n-1}^{(d,m)}(k) = 0 \\ \text{Beta}(e_0^1, f_0^1) & \text{if } z_{n-1}^{(d,m)}(k) = 1 \end{cases} \quad (5)$$

In our experiments, we let $e_0^1 = f_0^0 = 0.9 \times K$ and $f_0^1 = e_0^0 = 0.1 \times K$, same as the corresponding priors used in Ding et al. (2011). Since K is usually a large integer, $\mathbb{E}[u_{k,n}^{(d,m)}] = 0.1$ with a very small variance $\mathbb{C}[u_{k,n}^{(d,m)}]$ when $z_{n-1}^{(d,m)}(k) = 0$ and $\mathbb{E}[u_{k,n}^{(d,m)}] = 0.9$ with a very small

variance $\mathbb{C}[u_{k,n}^{(d,m)}]$ when $z_{n-1}^{(d,m)}(k) = 1$. Thus the sparseness pattern propagates across samples from a given aspect-frame with the high probability. There are many approaches to describe the temporal or spatial dependency via Bayesian frameworks (Chatzis & Tsechpenakis, 2010; Ding et al., 2011; Du, Ren, Dunson, & Carin, 2009, 2011; Ren, Dunson, & Carin, 2008), of which the Markov structure is much simpler and more suitable for our problem. For the HRR samples from an aspect-frame, the correlation dependencies in different range cells $\{k\}$ and between different neighboring samples $\{n-1, n\}$ may be quite different, since there are different types of dominant scatterers in each range cell and the dominant scatterers may migrate among the close range cells, especially for the superresolution case. As shown in (5), the different correlation dependency in each range cell k and for each sample n can be easily described via the simple Markov structure, which will not result in overfitting in the model learning.

The complex echo from a range cell is the summation of the complex-valued scattering responses from all scatterers in this range cell. According to the central limit theorem, the summation of many independent variables approximately follows the Gaussian distribution. Therefore, for the sake of simplicity, it is usually reasonable to assume that the real and imaginary parts of the complex echo from a range cell respectively follow the Gaussian distributions, and the complex-valued variable follows the complex Gaussian distribution. In fact, as shown in Fig. 1, the histograms of the inphase and quadrature components of the complex echoes from the measured data also approximate to Gaussian distributions, which means this assumption can work well for the measured HRR complex data. Then here $\hat{\mathbf{w}}_n^{(d,m)}$ is modeled as

$$\hat{\mathbf{w}}_n^{(d,m)} \sim \text{CNorm}(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0) \quad (6)$$

where $\text{CNorm}(\cdot)$ represents the complex Gaussian distribution.

$$p(\hat{\mathbf{w}}_n^{(d,m)} | \boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0) = \frac{1}{(\pi)^K |\boldsymbol{\Sigma}_0|} \times \exp \left\{ -(\hat{\mathbf{w}}_n^{(d,m)} - \boldsymbol{\mu}_0)^H \boldsymbol{\Sigma}_0^{-1} (\hat{\mathbf{w}}_n^{(d,m)} - \boldsymbol{\mu}_0) \right\} \quad (7)$$

We use the complex Gaussian distribution with the zero-mean ($\boldsymbol{\mu}_0 = \mathbf{0}$) and identity-covariance ($\boldsymbol{\Sigma}_0 = \mathbf{I}_K$) as the prior for $\hat{\mathbf{w}}_n^{(d,m)}$. The posterior mean and covariance parameters can be further inferred for each sample, which will be discussed in detail in Section 2.2.

2.1.2. Noise components

For aircraft-like targets under a look-up scenario, the noises in the inphase and quadrature echoes of targets can be assumed to be Gaussian white noises. Therefore, we assume the model noise is drawn *i.i.d.* from a complex Gaussian distribution with the zero-mean. The noise variance or precision is assumed unknown, and is learned within the model inference. Mathematically,

$$\eta_n^{(d,m)}(l) \sim \text{CNorm}(0, \alpha^{-1}), \quad \alpha \sim \text{Gamma}(a_0, b_0) \quad (8)$$

where $p(\eta_n^{(d,m)}(l) | \alpha) = (\pi \alpha^{-1})^{-1} \exp(-\alpha |\eta_n^{(d,m)}(l)|^2)$, and $\text{Gamma}(\cdot)$ represents the Gamma distribution with hyperparameters $a_0 > 0$ and $b_0 > 0$. We usually set $a_0 = b_0 = 10^{-6}$ (Ji, Xue, & Carin, 2008).

There are also some other Bayesian models for the linear regression problem $\mathbf{y} = \Phi \mathbf{w} + \boldsymbol{\eta}$ with the different sparse priors (Ding et al., 2011; Ji et al., 2008; Zhang et al., 2010). Zhang et al. (2010) uses Laplace prior on $\mathbf{w}$, and its maximum *a posteriori* (MAP) estimation can be derived. But since the Laplace prior is not conjugate to the Gaussian likelihood, an estimate of the full posterior on $\mathbf{w}$ cannot be readily accomplished. Therefore, Ji et al. (2008) defines a Gaussian-Gamma hierarchical prior on each element of the real-valued $\mathbf{w}$, which has the similar properties as the Laplace prior but allows the convenient conjugate-exponential analysis to derive the closed-form Bayesian inference for $\mathbf{w}$. Nevertheless, for Ji et al. (2008), it is hard to exploit the structure information (beyond just sparseness) in $\mathbf{w}$ with just one variable to represent both the coefficient values and the sparseness. Ding et al. (2011) proposes an equivalent form to separate the learning of sparseness from the learning of coefficient values as $\mathbf{w} = \hat{\mathbf{w}} \circ \mathbf{z}$, where $\mathbf{z}$ is a zero/one indicator vector and the Markov structure with Bernoulli-Beta prior can be easily imposed on it. The model in Ding et al. (2011) is proposed for the real-valued image CS analysis. In this paper, we propose a similar model structure to characterize the HRR complex signal and apply the signal model to target recognition problem.

2.2. Posterior computation

The posterior distributions of the model parameters $\boldsymbol{\Psi}$ are inferred from the observed training data $\mathbf{Y}$, via the VB inference (Beal, 2003), which converges fast and is computationally efficient. In VB inference, we seek a set of distributions $q(\boldsymbol{\Psi}) = \prod_h q_h(\boldsymbol{\Psi}_h)$ to approximate the true posterior distributions $p(\boldsymbol{\Psi} | \mathbf{Y})$, and use a

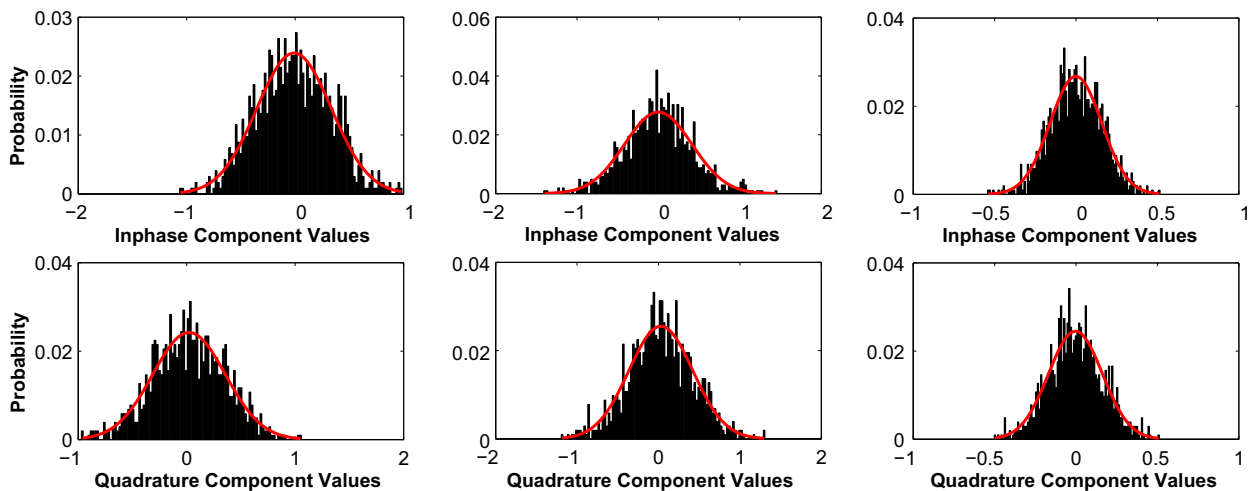


Fig. 1. Distribution examples of the inphase and quadrature components of the complex echoes from some range cells of Cessna Citation S/II data. The bars represent the histograms, and the red lines represent the Gaussian pdf with the mean and variance determined by the measured data. Each column is a pair of the inphase and quadrature components.

lower bound $\mathcal{L}(q(\Psi))$ to approximate the true log-likelihood of the model $\log p(\mathbf{Y} | \Psi)$. The algorithm iteratively updates $q(\Psi)$ so that $\mathcal{L}(q(\Psi))$ approaches to $\log p(\mathbf{Y} | \Psi)$ until convergence. In the following experiments, we terminate the algorithm when the change in $\mathcal{L}(q(\Psi))$ is negligibly small.

In our model $\Psi = \left\{ \left\{ \mathbf{z}_n^{(d,m)}, \hat{\mathbf{w}}_n^{(d,m)}, \left\{ \mathbf{u}_{k,n}^{(d,m)} \right\}_{k=1,n=1}^{K,N} \right\}_{d=1,m=1,n=1}^{D,M_d,N}, \alpha \right\}$ and $\mathbf{Y} = \left\{ \mathbf{y}_n^{(d,m)} \right\}_{d=1,m=1,n=1}^{D,M_d,N}$. The update equations for the posterior distributions of Ψ are as follows:

$$q(\mathbf{z}_n^{(d,m)}) = \prod_{k=1}^K \text{Ber}(z_n^{(d,m)}(k) | \rho_{k,n}^{(d,m)}), \quad q(\hat{\mathbf{w}}_n^{(d,m)}) = \text{CNorm}(\hat{\mathbf{w}}_n^{(d,m)} | \boldsymbol{\mu}_n^{(d,m)}, \boldsymbol{\Sigma}_n^{(d,m)})$$

$$q(\mathbf{u}_{k,n}^{(d,m)}) = \text{Beta}(u_{k,n}^{(d,m)} | e_{k,n}^{(d,m)}, f_{k,n}^{(d,m)}), \quad q(\alpha) = \text{Gamma}(\alpha | a, b) \quad (9)$$

with

$$\rho_{k,n}^{(d,m)} = \left[1 + \exp \left\{ \mathbb{E}[\ln(1 - u_{k,n}^{(d,m)})] - \mathbb{E}[\ln u_{k,n}^{(d,m)}] \right. \right. \\ \left. \left. + \mathbb{E}[\alpha] \left[\mathbb{E}[\hat{\mathbf{w}}_n^{(d,m)*}(k) \cdot \hat{\mathbf{w}}_n^{(d,m)}(k)] \phi(r_k)^H \phi(r_k) \right. \right. \right. \\ \left. \left. \left. - 2\mathbb{E}[\hat{\mathbf{w}}_n^{(d,m)}(k)]^* \phi(r_k)^H \mathbf{y}_n^{(d,m)(-k)} \right] \right\} \right]^{-1}$$

$$\mathbf{y}_n^{(d,m)(-k)} = \mathbf{y}_n^{(d,m)} - \sum_{k' \neq k} \phi(r_{k'}) \mathbb{E}[\mathbf{z}_n^{(d,m)}(k')] \cdot \mathbb{E}[\hat{\mathbf{w}}_n^{(d,m)}(k')]$$

$$\boldsymbol{\Sigma}_n^{(d,m)} = \left[\mathbf{I}_K + \mathbb{E}[(\Phi^H \circ \mathbf{Z}_n^{(d,m)})(\Phi \circ \mathbf{Z}_n^{(d,m)T})] \cdot \mathbb{E}[\alpha] \right]^{-1}$$

$$\boldsymbol{\mu}_n^{(d,m)} = \mathbb{E}[\alpha] \boldsymbol{\Sigma}_n^{(d,m)} \left[\mathbf{b}_{1,n}^T \mathbf{y}_n^{(d,m)(-1)}, \mathbf{b}_{2,n}^T \mathbf{y}_n^{(d,m)(-2)}, \dots, \mathbf{b}_{K,n}^T \mathbf{y}_n^{(d,m)(-K)} \right]^T$$

$$\mathbf{Z}_n^{(d,m)} = [\mathbf{z}_n^{(d,m)}, \mathbf{z}_n^{(d,m)}, \dots, \mathbf{z}_n^{(d,m)}] \in \mathbb{R}^{K \times L},$$

$$\mathbf{B}_n = [\mathbf{b}_{1,n}^T; \mathbf{b}_{2,n}^T; \dots; \mathbf{b}_{K,n}^T]^T = \Phi^H \mathbb{E}[\mathbf{Z}_n^{(d,m)}]$$

$$a = a_0 + L \times N \times \sum_{d=1}^D M_d$$

$$b = b_0 + \sum_{d=1}^D \sum_{m=1}^{M_d} \sum_{n=1}^N \left\{ \|\mathbf{y}_n^{(d,m)} - \Phi(\mathbb{E}[\mathbf{z}_n^{(d,m)}] \circ \mathbb{E}[\hat{\mathbf{w}}_n^{(d,m)}])\|_2^2 \right. \\ \left. + \text{trace} \left\{ \Phi^H \Phi \left[(\mathbb{E}[\hat{\mathbf{w}}_n^{(d,m)}] \hat{\mathbf{w}}_n^{(d,m)H}) \circ \mathbb{E}[\mathbf{z}_n^{(d,m)} \mathbf{z}_n^{(d,m)T}] \right] \right. \right. \\ \left. \left. - (\mathbb{E}[\hat{\mathbf{w}}_n^{(d,m)}] \circ \mathbb{E}[\mathbf{z}_n^{(d,m)}]) (\mathbb{E}[\hat{\mathbf{w}}_n^{(d,m)}] \circ \mathbb{E}[\mathbf{z}_n^{(d,m)}])^H \right\} \right\}$$

$$e_{(k,1)}^{(d,m)} = e_0 + \mathbb{E}[z_{k,1}^{(d,m)}], \quad f_{(k,1)}^{(d,m)} = f_0 + (1 - \mathbb{E}[z_{k,1}^{(d,m)}])$$

$$e_{(k,n)}^{(d,m)} = (1 - \mathbb{E}[z_{k,n-1}^{(d,m)}]) (e_0^0 + \mathbb{E}[z_{k,n}^{(d,m)}]) + \mathbb{E}[z_{k,n-1}^{(d,m)}] (e_0^1 + \mathbb{E}[z_{k,n}^{(d,m)}])$$

$$f_{(k,n)}^{(d,m)} = (1 - \mathbb{E}[z_{k,n-1}^{(d,m)}]) (f_0^0 + 1 - \mathbb{E}[z_{k,n}^{(d,m)}]) + \mathbb{E}[z_{k,n-1}^{(d,m)}] (f_0^1 + 1 - \mathbb{E}[z_{k,n}^{(d,m)}]) \quad (10)$$

where $\text{trace}\{\cdot\}$ represents to get the trace of the matrix in it, and

$$\mathbb{E}[z_n^{(d,m)}(k)] = \rho_{k,n}^{(d,m)}, \quad \mathbb{E}[z_n^{(d,m)2}(k)] = \rho_{k,n}^{(d,m)} \\ \mathbb{E}[z_n^{(d,m)}(k) \cdot z_n^{(d,m)}(k')] = \rho_{k,n}^{(d,m)} \cdot \rho_{k',n}^{(d,m)} \quad \text{with } k \neq k' \\ \mathbb{E}[\hat{\mathbf{w}}_n^{(d,m)}(k)] = \mu_n^{(d,m)}(k), \quad \mathbb{E}[\hat{\mathbf{w}}_n^{(d,m)*}(k) \cdot \hat{\mathbf{w}}_n^{(d,m)}(k)] \\ = \mu_n^{(d,m)*}(k) \cdot \mu_n^{(d,m)}(k) + \boldsymbol{\Sigma}_n^{(d,m)}(k, k) \\ \mathbb{E}[\alpha] = a/b, \quad \mathbb{E}[\ln u_{k,n}^{(d,m)}] = \psi(e_{k,n}^{(d,m)}) + \psi(e_{k,n}^{(d,m)} + f_{k,n}^{(d,m)} + 1) \\ \mathbb{E}[\ln(1 - u_{k,n}^{(d,m)})] = \psi(f_{k,n}^{(d,m)}) + \psi(e_{k,n}^{(d,m)} + f_{k,n}^{(d,m)} + 1) \quad (11)$$

where $\psi(\cdot)$ is the Digamma function (Beal, 2003).

As shown in (10), $q(\mathbf{z}_n^{(d,m)})$, $q(\hat{\mathbf{w}}_n^{(d,m)})$ and $q(u_{k,n}^{(d,m)})$ are updated for each sample, and the update equation of b in $q(\alpha)$ is invariant to each sample's initial-phase due to the conjugate product calculation. Thus the initial-phase sensitivity problem can be avoided in the model learning procedure.

3. Statistical recognition and statistical compressive sensing

3.1. Statistical recognition based on Bayesian model

According to Bayes' theorem, the posterior probability of the d th class ($d \in \{1, 2, \dots, D\}$ denotes class membership) $p(d | \mathbf{y}^*) \propto p(\mathbf{y}^* | d) p(d)$, where $p(\mathbf{y}^* | d)$ is the class-conditional likelihood and $p(d)$ is the prior class probability. Since the prior class probability is usually assumed to be uniformly distributed, estimation of the posterior probability is turned into estimation of the class-conditional likelihood. Furthermore, considering the target-aspect sensitivity of HRR data, we need to develop statistical model for each aspect-frame, and the class-conditional likelihood of each class will be obtained via substituting the test sample into all of the target-aspect dependent statistical models and searching the maximum one, i.e., $p(\mathbf{y}^* | d) = \max_m p(\mathbf{y}^* | d, m)$ with $p(\mathbf{y}^* | d, m)$ denoting the statistical aspect-frame template of the m th frame from the d th target ($m \in \{1, 2, \dots, M_d\}$ denotes the frame membership). Therefore, the main task in the training stage of statistical recognition is to estimate each statistical aspect-frame template of each target.

3.1.1. Training stage

After the signal model is inferred for each sample from a given aspect-frame as discussed in Section 2.2, the corresponding statistical frame model can be estimated as follows

$$p(\mathbf{y} | d, m) = \iint \text{CNorm}(\mathbf{y}; \Phi \mathbf{w}^{(d,m)}, \alpha^{-1} \mathbf{I}_L) \times \text{Gamma}(\alpha) \\ \times p(\mathbf{w}^{(d,m)}) d\alpha d\mathbf{w} \quad (12)$$

where $\text{Gamma}(\alpha)$ represents the posterior Gamma distribution of α , $p(\mathbf{w}^{(d,m)})$ denotes the posterior pdf for all HRR complex samples from the m th frame of the d th target.

According to the signal model, for each sample in the aspect-frame, $\mathbf{w}_n^{(d,m)} = \hat{\mathbf{w}}_n^{(d,m)} \circ \mathbf{z}_n^{(d,m)}$. As shown in the posterior computation in Section 2.2, the posterior distribution of $\hat{\mathbf{w}}_n^{(d,m)}$ depends on the posterior mean of $\mathbf{z}_n^{(d,m)}$, which is binary when the inference converges, thus the posterior distribution $p(\mathbf{w}_n^{(d,m)})$ approximately equals to the posterior distribution $p(\hat{\mathbf{w}}_n^{(d,m)})$. Since the scattering coefficient vector of each sample in the aspect-frame follows a complex Gaussian distribution, we can assume that the posterior pdf $p(\mathbf{w}^{(d,m)})$ is also a complex Gaussian distribution:

$$\mathbf{w}^{(d,m)} \sim \text{CNorm}(\boldsymbol{\mu}^{(d,m)}, \boldsymbol{\Sigma}^{(d,m)}) \\ \boldsymbol{\mu}^{(d,m)} = \frac{1}{N} \sum_{n=1}^N \mathbb{E}[\hat{\mathbf{w}}_n^{(d,m)}] \\ \boldsymbol{\Sigma}^{(d,m)} = \frac{1}{N} \sum_{n=1}^N \left\{ (\mathbb{E}[\hat{\mathbf{w}}_n^{(d,m)}] - \boldsymbol{\mu}^{(d,m)}) (\mathbb{E}[\hat{\mathbf{w}}_n^{(d,m)}] - \boldsymbol{\mu}^{(d,m)})^H \right\} \quad (13)$$

where $\mathbb{E}[\hat{\mathbf{w}}_n^{(d,m)}]$ represents the posterior expectation of $\hat{\mathbf{w}}_n^{(d,m)}$. Paper (Chen et al., 2010) used the similar estimator from Gibbs sampling for deriving the CS inversion for real-valued image data.

The phase term of the k th dominant scatterer in the n th sample $\theta_k(n) = -\frac{4\pi}{\lambda} r_k(n)$, where λ denotes the radar wavelength and $r_k(n)$ denotes the radial distance from this scatterer to the radar. If we assume that the radial distances of the most scatterers in a target $\{r_k(n)\}_k$ uniformly vary and the corresponding phase terms

$\{\theta_k(n)\}_k$ are uniform-distributed within $[0, 2\pi]$ across the samples within the aspect-frame (i.e., for $n = 1, 2, \dots, N$), then the mean value of the complex echoes from each of these scatterers within the aspect-frame must approximate to zero. In this sense, it is reasonable to assume that $\mathbf{w}^{(d,m)}$ follows a complex Gaussian distribution with zero-mean. Therefore, for our application, (13) can be simplified as

$$\mathbf{w}^{(d,m)} \sim \text{CNorm}(\mathbf{0}, \Sigma^{(d,m)}), \quad \Sigma^{(d,m)} = \frac{1}{N} \sum_{n=1}^N \mathbb{E}[\hat{\mathbf{w}}_n^{(d,m)}] \mathbb{E}[\hat{\mathbf{w}}_n^{(d,m)}]^H \quad (14)$$

Then due to the conjugacy, we can calculate the integrals in (12) and get the frame model

$$p(\mathbf{y}|d, m) = \text{CNorm}(\mathbf{0}, \Lambda^{(d,m)}) \quad (15)$$

$$\Lambda^{(d,m)} = \Phi \left(\frac{1}{N} \sum_{n=1}^N \mathbb{E}[\hat{\mathbf{w}}_n^{(d,m)}] \mathbb{E}[\hat{\mathbf{w}}_n^{(d,m)}]^H \right) \Phi^H + (\mathbb{E}[\alpha])^{-1} \mathbf{I}_L$$

where the posterior expectations $\{\mathbb{E}[\hat{\mathbf{w}}_n^{(d,m)}]\}_{n=1}^N$ and $\mathbb{E}[\alpha]$ can be easily got after the signal model inference.

3.1.2. Classification stage

If the frequency response of a test HRR complex sample, i.e., $\mathbf{y}^*$, is measured under the same SNR circumstance as that of training data, its frame-conditional likelihood given the frame m of the target d can be calculated according to (15). Based on the property of complex Gaussian distribution,

$$p(\mathbf{y}^*|d, m) = \text{CNorm}(\mathbf{y}^*; \mathbf{0}, \Lambda^{(d,m)})$$

$$= \text{Norm} \left(\begin{bmatrix} \text{Re}(\mathbf{y}^*) \\ \text{Im}(\mathbf{y}^*) \end{bmatrix}; \begin{bmatrix} \mathbf{0} \\ \mathbf{0} \end{bmatrix}, \frac{1}{2} \begin{bmatrix} \text{Re}(\Lambda^{(d,m)}) & -\text{Im}(\Lambda^{(d,m)}) \\ \text{Im}(\Lambda^{(d,m)}) & \text{Re}(\Lambda^{(d,m)}) \end{bmatrix} \right) \quad (16)$$

Then the class-conditional likelihood of target d is $p(\mathbf{y}^*|d) = \max_m p(\mathbf{y}^*|d, m)$, and its posterior probability is $p(d|\mathbf{y}^*) = \frac{p(\mathbf{y}^*|d)p(d)}{\sum_{d=1}^C p(\mathbf{y}^*|d)p(d)}$. The class membership of $\mathbf{y}^*$ should be assigned to the class with the maximum posterior probability, i.e., $d^* = \arg \max_d p(d|\mathbf{y}^*)$. Similar frame matching method is also utilized in Du et al. (2006, 2008b, 2011, 2012).

For the RATR problem, the training data can usually be obtained under high SNR and without any interference via some cooperative measurement experiments or directly via simulations; while the test samples are usually got in the non-cooperative circumstance (e.g., at the battle time), where the high SNR and no interference condition cannot be guaranteed due to the long distance away from the non-cooperative targets and some interference signals from the measurement system or environment (e.g., some communication signals with the frequencies within our radar band). Please note here we only consider the accidental narrowband interference problem (not the cheating interference in the electronic warfare), i.e., a small part of frequency components of a test HRR complex sample $\{\bar{\mathbf{y}}^*(l)\}_{l=l_1}^{l_2}$ with $l_2 - l_1 + 1 < L$ are impacted by some interference signals with the same frequencies. Here we assume that the measurement $\bar{\mathbf{y}}^* = [\bar{\mathbf{y}}^*(1), \bar{\mathbf{y}}^*(2), \dots, \bar{\mathbf{y}}^*(l_1 - 1), \bar{\mathbf{y}}^*(l_2 + 1), \bar{\mathbf{y}}^*(l_2 + 2), \dots, \bar{\mathbf{y}}^*(L)]^T \in \mathbb{C}^{L \times 1}$ is obtained via a projection $\bar{\mathbf{y}}^* = \mathbf{A}\mathbf{y}^* + \mathbf{v}^*$, where $\mathbf{A} \in \{0, 1\}^{L_1 \times L}$ is a binary projection matrix with $L_1 = L - (l_2 - l_1 + 1)$, $L_1 < L$ and $\mathbf{A}\mathbf{A}^T = \mathbf{I}_{L_1}$, $\mathbf{y}^*$ represents the original test sample in the full band and without the measurement noise, $\mathbf{v}^*$ denotes the new noise component introduced from the test condition with the assumption that $\mathbf{v}^* \sim \text{CNorm}(\mathbf{0}, r_*^{-1} \mathbf{I}_{L_1})$. If there's no interference, $\mathbf{A} = \mathbf{I}_L$. Then, based on the above linear projection structure, we can get the modified statistical model for the noised and interfered test sample is as follows:

$$p(\bar{\mathbf{y}}^*|d, m) = \int \text{CNorm}(\mathbf{A}\mathbf{y}^*, r_*^{-1} \mathbf{I}_{L_1}) \times p(\mathbf{y}^*|d, m) d\mathbf{y}^*$$

$$= \text{CNorm}(\bar{\mathbf{y}}^*; \mathbf{0}, \mathbf{A}\Lambda^{(d,m)}\mathbf{A}^T + r_*^{-1} \mathbf{I}_{L_1}) \quad (17)$$

There are several ways to accurately estimate the noise power (i.e., noise variance r_*^{-1}) and the band of the accidental interference signal (i.e., $[l_1, l_2]$), of which a simple approach is to measure the noise power and signal bandwidth via the radar receiver with no input signal, namely a passive system. Therefore, the matched target indicator d^* and frame indicator m^* can be easily obtained via the above frame matching method based on $\{p(\bar{\mathbf{y}}^*|d, m)\}_{m=1, d=1}^{M_d, D}$.

The precondition for the above derivation in (17) is the linear signal model $\bar{\mathbf{y}}^* = \mathbf{A}\mathbf{y}^* + \mathbf{v}^*$ for the noised and interfered HRR complex data, which will not hold for the HRRP data or the FFT-magnitude feature of the HRR complex data due to the absolute value operator. Thus here we should restate again the linear signal model for the HRR complex data under the low SNR or with the narrowband interference is an obvious advantage of statistical recognition of the HRR complex data over that of the real-valued features.

3.2. Statistical CS inversion

The traditional CS methods can recover the compressible, noisy signals through the solution of the computationally tractable sparse constrained inverse problem. For the CS-based radar imaging application, Cetin and Karl (2001) proposes a feature enhancement algorithm for radar imaging via solving a sparse linear regression problem based on nonquadratic regularization (Zhang et al., 2010) proposes a CS imaging method based on Bayesian sparse learning; (Patel et al., 2010) proposes two schemes to realize the random projection for radar imaging. Nevertheless, all these papers only utilize the underlying sparse property of a set of HRR signals corresponding to an image to achieve the recovery of the radar image.

If $p(\mathbf{y}^*)$ is known via statistical learning, for the projection measurement $\bar{\mathbf{y}}^* = \mathbf{A}\mathbf{y}^* + \mathbf{v}^*$, the condition distribution of $\mathbf{y}^*$ given $\bar{\mathbf{y}}^*$ is

$$p(\mathbf{y}^*|\bar{\mathbf{y}}^*) = \frac{p(\mathbf{y}^*)p(\bar{\mathbf{y}}^*|\mathbf{y}^*)}{p(\bar{\mathbf{y}}^*)} = \frac{p(\mathbf{y}^*)p(\bar{\mathbf{y}}^*|\mathbf{y}^*)}{\int p(\mathbf{y}^*)p(\bar{\mathbf{y}}^*|\mathbf{y}^*)d\mathbf{y}^*} \quad (18)$$

Then the mean squared error between the recovery signal $\hat{\mathbf{y}}^*$ and the unknown signal $\mathbf{y}^*$ is

$$J = \mathbb{E}_{\mathbf{y}^*|\bar{\mathbf{y}}^*} [\|\mathbf{y}^* - \hat{\mathbf{y}}^*\|_2^2] = \int \|\mathbf{y}^* - \hat{\mathbf{y}}^*\|_2^2 p(\mathbf{y}^*|\bar{\mathbf{y}}^*) d\mathbf{y}^* \quad (19)$$

where $\mathbb{E}_{\mathbf{y}^*|\bar{\mathbf{y}}^*}[\cdot]$ means the expectation with respect to $\mathbf{y}^*$ and with $\bar{\mathbf{y}}^*$ known. Then the optimal recovery with minimum mean squared error (MMSE) should be

$$\hat{\mathbf{y}}^* = \mathbb{E}_{\mathbf{y}^*|\bar{\mathbf{y}}^*}[\mathbf{y}^*] \quad (20)$$

The above derivation is under the assumption that $p(\mathbf{y}^*)$ is known as the prior information. After the class membership d^* and the frame membership m^* of $\bar{\mathbf{y}}^*$ are determined via the statistical recognition method discussed above, $p(\mathbf{y}^*) = p(\mathbf{y}^*|d^*, m^*)$. Then $p(\mathbf{y}^*|\bar{\mathbf{y}}^*) = p(\mathbf{y}^*|\bar{\mathbf{y}}^*, d^*, m^*)$ may be analytically calculated

$$p(\mathbf{y}^*|\bar{\mathbf{y}}^*, d^*, m^*) = \frac{p(\mathbf{y}^*|d^*, m^*)p(\bar{\mathbf{y}}^*|\mathbf{y}^*)}{\int p(\mathbf{y}^*|d^*, m^*)p(\bar{\mathbf{y}}^*|\mathbf{y}^*)d\mathbf{y}^*}$$

$$= \frac{\text{CNorm}(\mathbf{y}^*; \mathbf{0}, \Lambda^{(d^*, m^*)}) \times \text{CNorm}(\bar{\mathbf{y}}^*; \mathbf{A}\mathbf{y}^*, r_*^{-1} \mathbf{I}_{L_1})}{\int \text{CNorm}(\mathbf{y}^*; \mathbf{0}, \Lambda^{(d^*, m^*)}) \times \text{CNorm}(\bar{\mathbf{y}}^*; \mathbf{A}\mathbf{y}^*, r_*^{-1} \mathbf{I}_{L_1}) d\mathbf{y}^*}$$

$$= \frac{\text{CNorm}(\mathbf{y}^*; \tilde{\boldsymbol{\mu}}^{(d^*, m^*)}, \tilde{\boldsymbol{\Lambda}}^{(d^*, m^*)}) \times \text{CNorm}(\bar{\mathbf{y}}^*; \mathbf{0}, \mathbf{A}\Lambda^{(d^*, m^*)}\mathbf{A}^T + r_*^{-1} \mathbf{I}_{L_1})}{\int \text{CNorm}(\mathbf{y}^*; \tilde{\boldsymbol{\mu}}^{(d^*, m^*)}, \tilde{\boldsymbol{\Lambda}}^{(d^*, m^*)}) d\mathbf{y}^* \times \text{CNorm}(\bar{\mathbf{y}}^*; \mathbf{0}, \mathbf{A}\Lambda^{(d^*, m^*)}\mathbf{A}^T + r_*^{-1} \mathbf{I}_{L_1})}$$

$$= \text{CNorm}(\mathbf{y}^*; \tilde{\boldsymbol{\mu}}^{(d^*, m^*)}, \tilde{\boldsymbol{\Lambda}}^{(d^*, m^*)}) \quad (21)$$

with

$$\begin{aligned}\tilde{\mathbf{A}}^{(d^*, m^*)} &= \left(\mathbf{A}^{(d^*, m^*)^{-1}} + \mathbf{A}^T (r_* \mathbf{I}_{L_1}) \mathbf{A} \right)^{-1} \\ &= \mathbf{A}^{(d^*, m^*)} - \mathbf{A}^{(d^*, m^*)} \mathbf{A}^T (\mathbf{A} \mathbf{A}^{(d^*, m^*)} \mathbf{A}^T + r_*^{-1} \mathbf{I}_{L_1})^{-1} \mathbf{A} \mathbf{A}^{(d^*, m^*)} \\ \tilde{\boldsymbol{\mu}}^{(d^*, m^*)} &= \tilde{\mathbf{A}}^{(d^*, m^*)} \left(\mathbf{A}^T (r_* \mathbf{I}_{L_1}) \tilde{\mathbf{y}}^* \right) \\ &= \mathbf{A}^{(d^*, m^*)} \mathbf{A}^T (\mathbf{A} \mathbf{A}^{(d^*, m^*)} \mathbf{A}^T + r_*^{-1} \mathbf{I}_{L_1})^{-1} \tilde{\mathbf{y}}^*\end{aligned}\quad (22)$$

where d^* and m^* denote the matched target indicator and frame indicator. Since we have the closed-form expressions for all the parameters in the posterior distribution $p(\mathbf{y}^* | \tilde{\mathbf{y}}^*, d^*, m^*)$, according to (20), $\mathbb{E}_{\mathbf{y}^* | \tilde{\mathbf{y}}^*}[\mathbf{y}^*] = \tilde{\boldsymbol{\mu}}^{(d^*, m^*)}$ is the CS inversion for $\tilde{\mathbf{y}}^* \rightarrow \mathbf{y}^*$.

Please note here we consider a fixed projection matrix $\mathbf{A}$ in the CS problem. CS methods usually select a random matrix as the projection matrix, since the restricted isometry property (RIP) condition can be achieved with high probability via a random projection matrix (Candes & Tao, 2007). Nevertheless, sometimes we are interested in some applications with the fixed projection matrixes, e.g., superresolution SAR/ISAR imaging in cross-range dimension and suppression of narrowband interference for interfered target signals. As shown in Zhang et al. (2010) and Cetin and Karl (2001), although the RIP condition may not be satisfied via the fixed projection matrix for the superresolution imaging application, the CS superresolution methods can also achieve some improvement in the performance. In addition, our statistical CS inversion utilizes the statistical information learned in the training stage beyond just sparseness, which can guarantee the better reconstruction performance.

4. Experimental results

4.1. Measured data

We examine the performance of the proposed method on the measured HRR data from the three real airplanes, which are also used in Du et al. (2005, 2006, 2008b, 2011, 2012) and Shi et al. (2011). The parameters of the airplane targets and measurement radar are shown in Table 1, and the projections of target trajectories onto the ground plane are shown in Fig. 2, from which the aspect angle of an airplane can be estimated according to its relative position to radar. As shown in Fig. 2, the measured data are segmented, where each segment contains 25,600 HRRP

Table 1
Parameters of planes and radar in the measurement experiment.

Radar parameters	Center frequency: 5520 MHz Bandwidth: 400 MHz		
Planes	Length (m)	Width (m)	Height (m)
Yak-42	36.38	34.88	9.83
Cessna Citation S/II	14.40	15.90	4.57
An-26	23.80	29.20	9.83

samples and each sample is a 256-length vector, i.e., $L = 256$. In our recognition experiments, data from the 2nd segment and a part of the 5th segment of Yak-42, the 6th and the 7th segments of Cessna Citation S/II, the 5th and the 6th segments of An-26 are taken as the training samples, and other data segments are taken as test samples. In the following recognition and reconstruction experiments, the recognition accuracies and reconstruction errors are averaged over all of the test samples from the three targets.

This HRR dataset is measured under high SNR (≈ 40 dB) and without any interference. In order to evaluate the noise robustness of our methods, we add simulated white noises to the inphase and quadrature components of the complex test samples. The SNR is defined as

$$\text{SNR} = 10 \times \log_{10} \left(\frac{\sum_{l=1}^L P_{y_l}}{L \times P_{\text{Noise}}} \right) \quad (23)$$

where P_{y_l} denotes the power of the l th frequency response from the original test sample, L denotes the number of frequency components (here $L = 256$), and P_{Noise} denotes the power of noise in each frequency component. In addition, we only use the part of frequency components without interference to distinguish a interfered test sample, i.e., $\tilde{\mathbf{y}}^*$ instead of $\mathbf{y}^*$ as stated in Section 3.1.2.

4.2. Data preprocessing

It is a prerequisite for HRR radar target recognition to deal with the target-aspect, time-shift and amplitude-scale sensitivity problems (Du et al., 2005, 2006).

We have discussed the target-aspect sensitivity issue in Section 2.1. Same as those in our previous work (Du et al., 2005, 2006, 2008b, 2012; Shi et al., 2011), we use the aspect-frame partition method to deal with the target-aspect sensitivity of the training dataset. There are 35 aspect-frames for Yak-42, 50 for Cessna Citation S/II, and 50 for An-26, of which each aspect-frame totally has 1024 HRRP samples, i.e., $D = 3$, $M_1 = 35$, $M_2 = M_3 = 50$, $N = 1024$.

Since an HRR sample is only a part of the received radar echo extracted by a range window, in which target signal is included, the position of the target signal in an HRR sample can vary with the measurement. This is the time-shift sensitivity. A time-shift on an HRR complex sample is transformed to a linear phase term on its complex frequency response. In this paper, we construct the statistical model for the complex frequency responses from HRR complex samples. In the training stage, the traditional motion compensation method, e.g., the slide correlation processing (Walker, 1980), is used for the HRR complex samples from each aspect-frame, and the corresponding complex frequency responses from the time-shift compensated (i.e. motion compensated) HRR complex data are utilized in the following model learning. In the classification stage, we use the linear phase terms with different time-shifts to compensate the test complex frequency response,

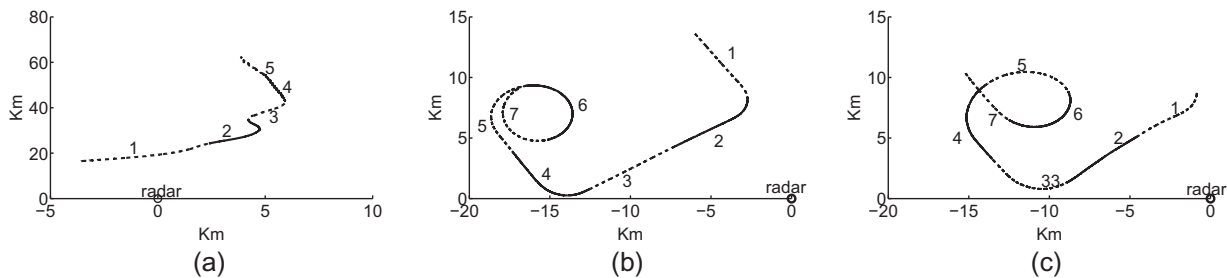


Fig. 2. Projections of target trajectories onto the ground plane: (a) Yak-42; (b) Cessna Citation S/II; (c) An-26.

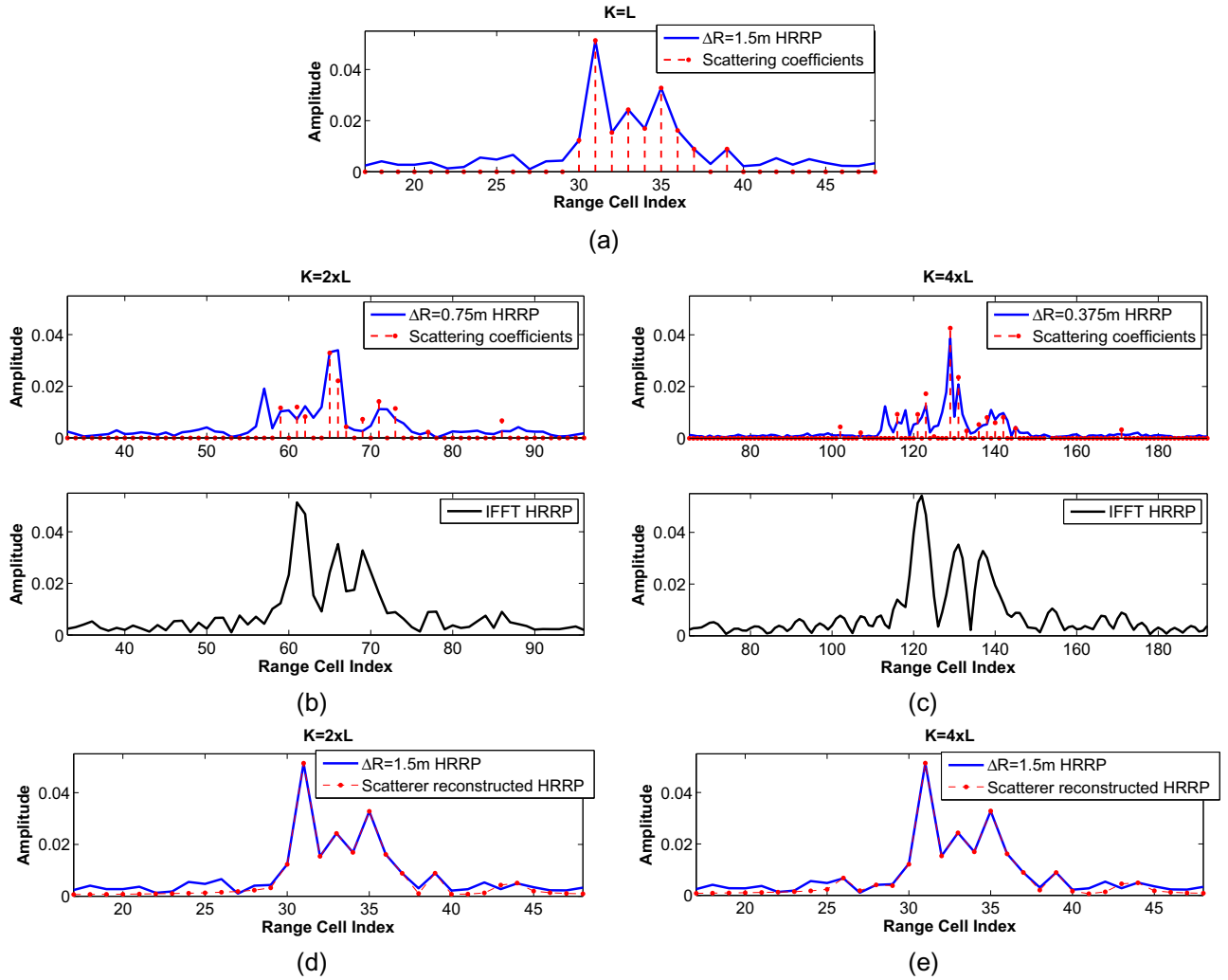


Fig. 3. A signal model learning example from Cessna Citation S/II HRR data. The signal model learning and reconstructions use 1.5 m resolution data, i.e., $L = 64$. (a) Reference profile at 1.5 m resolution and the learned amplitudes of the scattering coefficients via the Bayesian model with $K = L$. (b) Reference profile at 0.75 m resolution, the learned amplitudes of the scattering coefficients via the Bayesian model with $K = 2 \times L$ and the superresolution HRRP via IFFT with $K = 2 \times L$. (c) Reference profile at 0.375 m resolution, the learned amplitudes of the scattering coefficients via the Bayesian model with $K = 4 \times L$ and the superresolution HRRP via IFFT with $K = 4 \times L$. (d) Reference profile at 1.5 m resolution and the reconstructed 1.5 m HRRP via the Bayesian model with $K = 2 \times L$. (e) Reference profile at 1.5 m resolution and the reconstructed 1.5 m HRRP via the Bayesian model with $K = 4 \times L$.

and select that with the maximum likelihood as the optimal compensated test sample for the following recognition decision.

Furthermore, since the intensity of an HRR sample is a function of radar transmitting power, target distance, radar antenna gain, radar receiver gain, radar system losses and so on, HRR samples measured by different radars or under different conditions will have different amplitude-scales. To deal with this amplitude-scale sensitivity, each complex frequency response sample is normalized by dividing its l_2 -norm.

4.3. Signal model analysis

To illustrate the accuracy of the parameter estimation of our Bayesian model, a simulation experiment is performed in the following. According to Table 1, the range resolution of our measured data is around 0.375 m, since $\Delta R \approx \frac{c}{2B}$ with ΔR denoting the range resolution, $c = 3.0 \times 10^8$ m/s denoting the propagation velocity of radar signal and B denoting the signal bandwidth. In the following experiment, the range resolution was decreased from 0.375 m to 1.5 m (i.e., $L = 64$) to test our signal model, and the 0.375 m (i.e.,

$L = 256$) or 0.75 m (i.e., $L = 128$) HRRP samples were used as the “reference” profiles.

Fig. 3 shows some examples from Cessna Citation S/II data. Fig. 3(a)–(c) give the learned amplitudes of the scattering coefficients via the Bayesian model with $K = L$, $K = 2 \times L$ and $K = 4 \times L$, and the superresolution HRRPs via the conventional Fourier spectrum method with $K = 2 \times L$ and $K = 4 \times L$, i.e., the inverse fast Fourier transformation (IFFT) superresolution HRRPs, shown in the bottom subfigures of Fig. 3(b) and (c). As shown in Fig. 3(a), our signal model can learn the locations of the dominant scatterers, and their learned amplitudes are exactly same as those in the 1.5 m HRRP, since Φ is a complete orthogonal basis matrix with $K = L$. For the superresolution cases, i.e., Fig. 3(b) and (c), since Φ is redundant and some real scattering centers in the target may not be at the locations corresponding to the columns (atoms) in Φ , the learned amplitudes of the scattering coefficients are not exactly same as those in the 0.75 m HRRP or 0.375 m HRRP, which is shown in the blue line. Nevertheless, compared with the evident resolution loss of the conventional Fourier spectrum method, where several peaks of the reference HRRP have been merged into one in the IFFT superresolution HRRP with $K = 2 \times L$ or $K = 4 \times L$, our signal model not only has the better performance in terms of

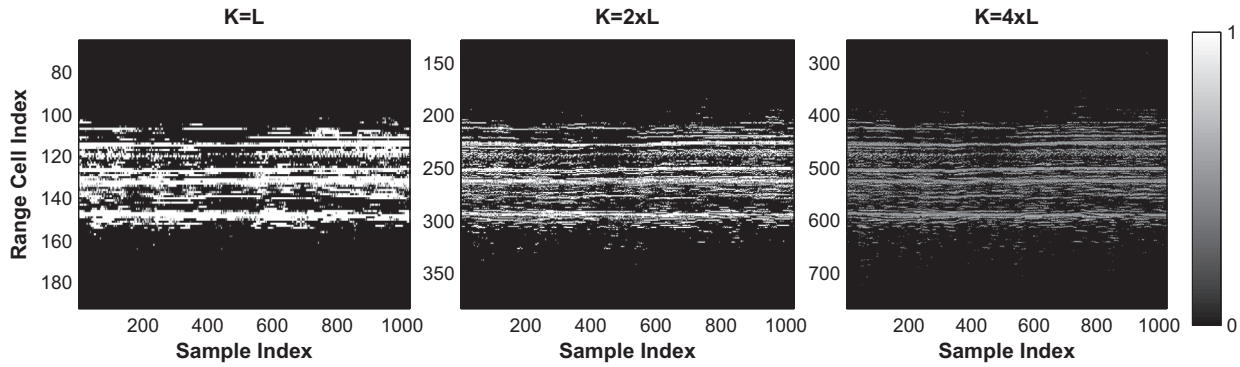


Fig. 4. Learned locations of the dominant scatterers for the 0.375 m HRR data (i.e., $L = 256$) from the first aspect-frame of Yak-42, i.e., $\mathbf{Z}^{(1,1)} = [\mathbb{E}[\mathbf{z}_1^{(1,1)}], \mathbb{E}[\mathbf{z}_2^{(1,1)}], \dots, \mathbb{E}[\mathbf{z}_{1024}^{(1,1)}]]$, generated via the Bayesian models with $K = L$, $K = 2 \times L$ and $K = 4 \times L$.

resolving the scatterers, but also learn the profile more similar to the reference 0.75 m HRRP or 0.375 m HRRP, as shown in Fig. 3(b) and (c). Fig. 3(d) and (e) give the reconstructed 1.5 m HRRPs via our Bayesian model with $K = 2 \times L$ and $K = 4 \times L$, which are very similar to the reference 1.5 m HRRP. The larger K is, the more precise the reconstruction is. Fig. 3(d) and (e) further demonstrate the accuracy of the parameter estimation of our signal model.

Fig. 4 shows $\{\mathbb{E}[\mathbf{z}_n^{(1,1)}]\}_{n=1}^{1024}$ for the 0.375 m HRR data (i.e., $L = 256$) from Yak-42, which are learned via the Bayesian model with $K = L$, $K = 2 \times L$ and $K = 4 \times L$. For the n th sample from the m th aspect-frame of the d th target, if a dominant scatterer is located at the k th range cell, then the posterior probability on this range cell usage, $\mathbb{E}[\mathbf{z}_n^{(d,m)}(k)]$, should be one; otherwise, it will be zero. Each column in each subfigure of Fig. 4 denotes the inferred locations of the dominant scatterers for each sample in the frame. As shown in Fig. 4, the larger K is, the more precise superresolution we can achieve. Given the parameter K , most dominant scatterers from the temporally proximate HRR samples are located at the same range cells, and also the distant repetitions or “innovations” occur for some samples, which accords with our Markov structure assumption. In addition, the number of the dominant scattering centers of an HRR sample varies across the sample index, which justifies using the Bernoulli-Beta prior for the automatic model selection.

4.4. Recognition performance

Fig. 5(a) depicts the average recognition rates for the three air-plane targets versus SNR, generated via the eight statistical recognition methods, including the two versions of the noise-robust MTL-FA model (Du et al., 2012) (i.e., the noise-robust MTL-FA model with the test noise level prior and the noise-robust MTL-FA model with the adaptively learned test noise level), the Gaussian model (Jacobs, 1999) without denoising, the Gaussian model with the test samples denoised via the orthogonal matching pursuit (OMP) based denoising method (Li, Zhou, Zhao, & Quan, 2014) for the two cases with $K = L$ and $K = 4 \times L$, and the robust recognition methods based on our Bayesian models for the three cases with $K = L$, $K = 2 \times L$ and $K = 4 \times L$. All these recognition methods are trained with the training data under high SNR, but the classifications are implemented under different SNR. The noise-robust MTL-FA model and Bayesian model can update the statistical models for the noised test data in the classification stage. The sparse representation denoising method for HRR signals proposed in Li et al. (2014) is based on the greedy CS method, OMP (Tropp & Gilbert, 2007). Here we use it as the denoising preprocessing for the Gaussian model. Since there is no interference in

this experiment, then the projection matrix $\mathbf{A}$ shown in the left subfigure Fig. 5(a) is an identity matrix. The MTL-FA model is used for the FFT-magnitude feature extracted from the HRR complex data, the Gaussian model is for the real HRRP data, the sparse representation denoising method and our recognition method are directly for the complex frequency responses from the HRR complex data. For brevity in Fig. 5, we omit the recognition curves from the five traditional statistical models, i.e., Gamma model (Webb, 2000), the two-distribution compounded model (Du et al., 2006), the single task learning based FA (STL-FA) model (Du et al., 2008b), the dynamic MTL-HMM model (Du et al., 2011) and the MTL-FA model without modification in classification (Du et al., 2012). From Du et al. (2012) and Du et al. (2011), the noise-robust MTL-FA models outperform the five models under the condition of low SNR.

As shown in Fig. 5(a), our Bayesian model with $K = 4 \times L$ is superior to other seven methods under each SNR case; when $\text{SNR} \geq 35$ dB, the noise-robust MTL-FA model with the test noise level prior yields a little better recognition rates than our Bayesian model with $K = 2 \times L$, and the noise-robust MTL-FA model with the adaptively learned test noise level yields a little better recognition rates than our Bayesian model with $K = L$; when $\text{SNR} \leq 20$ dB, the seven noise-robust recognition methods (the noise-robust MTL-FA models, the Gaussian methods with denoising and the noise-robust methods based on Bayesian models) outperform the Gaussian model without denoising; when $\text{SNR} \leq 15$ dB (this condition can be usually met for the non-cooperative measurement experiments, according to our experience with a large amount of measured data), the recognition performances of the three Bayesian models are superior to the noise-robust MTL-FA models and the Gaussian methods with denoising. If we assume that the average recognition rate threshold for the real application is 80%, the noise-robust recognition methods based on Bayesian models can work under the test condition of $\text{SNR} \geq 5$ dB; while the noise-robust MTL-FA models and the Gaussian model with denoising ($K = 4 \times L$) need $\text{SNR} \geq 15$ dB. Thus, according to the radar equation, the 10 dB advantage in SNR via our new methods can bring an increase about 80% in the classification distance between the target and radar in the real application.

Fig. 5(b) depicts the average recognition rates of the noised and interfered test data versus SNR. The classifications are implemented under different SNR conditions and with the narrowband interference of $\{\bar{\mathbf{y}}^*(l)\}_{l=56}^{87}$ and $\{\bar{\mathbf{y}}^*(l)\}_{l=169}^{200}$ impacted by some interference signals. Here the recognition results are generated via the Gaussian model with the IFFT test HRRPs from the noised and incomplete complex frequency response with $\{\bar{\mathbf{y}}^*(l)\}_{l=56}^{87} = 0$ and $\{\bar{\mathbf{y}}^*(l)\}_{l=169}^{200} = 0$, the Gaussian model with the test samples

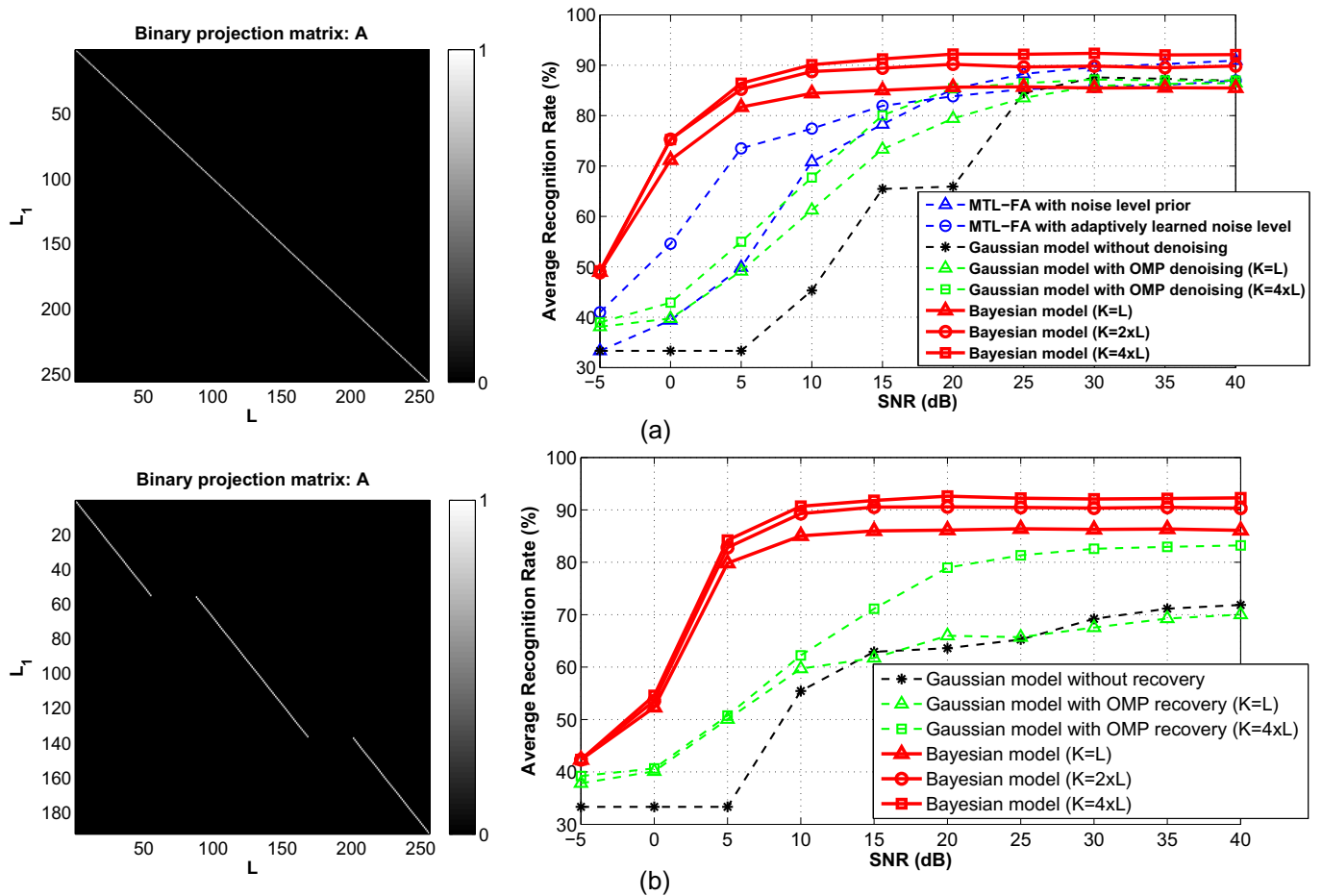


Fig. 5. Variation of the average recognition performance for the three airplane targets with SNR. (a) There is no interference, then the projection matrix A shown in the left subfigure is an identity matrix. Here we compare the recognition performances of the noise-robust MTL-FA model with the test noise level prior (Du et al., 2012); the noise-robust MTL-FA model with the adaptively learned test noise level (Du et al., 2012); the Gaussian model (Jacobs, 1999) without denoising; the Gaussian model with the test samples denoised via the OMP-based denoising method (Li et al., 2014) for the two cases with $K = L$ and $K = 4 \times L$; our robust recognition method based on Bayesian models for the three cases with $K = L$, $K = 2 \times L$ and $K = 4 \times L$. (b) The test samples are interfered via the projection matrix A shown in the left subfigure, where $\{\bar{y}^*(l)\}_{l=56}^{87}$ and $\{\bar{y}^*(l)\}_{l=169}^{200}$ are impacted by some interference signals. Here we compare the recognition performances of the Gaussian model with the IFFT test HRRPs from the noised and incomplete complex frequency responses with $\{\bar{y}^*(l)\}_{l=56}^{87} = 0$ and $\{\bar{y}^*(l)\}_{l=169}^{200} = 0$; the Gaussian model with the test samples reconstructed via OMP (Tropp & Gilbert, 2007) for the two cases with $K = L$ and $K = 4 \times L$; our robust recognition method based on Bayesian models for the three cases with $K = L$, $K = 2 \times L$ and $K = 4 \times L$. All these recognition methods are trained with the training data under high SNR and without any interference, but the classifications are implemented under different SNR and the subfigure (b) also with the narrowband interference.

reconstructed via OMP for the two cases with $K = L$ and $K = 4 \times L$, and the robust recognition methods based on our Bayesian models for the three cases with $K = L$, $K = 2 \times L$ and $K = 4 \times L$. As shown in Fig. 5(b), our Bayesian models are superior to other three methods under each SNR case. Compared with the results in Fig. 5(a), the recognition rates of the interfered test data are similar to those without any interference via our robust recognition methods when $\text{SNR} \geq 5$ dB, which demonstrates the statistical models updated via (17) and the interfered complex test data match well in the classification stage.

Here we should emphasize that the linear signal model for the HRR complex data and the corresponding updating of the statistical models according to (17) guarantee the better recognition performance of our method under low SNR or with narrowband interference.

4.5. CS reconstruction performance

As discussed in Section 3.2, the proposed statistical CS inversion can reconstruct the HRR complex sample without noise or interference from the noised or interfered sample. The reconstruction

performance quantified via the average relative reconstruction errors (i.e., root mean square errors (RMSEs)) is shown in Fig. 6. The RMSE is defined as $\frac{1}{N^*} \sum_{n=1}^{N^*} \frac{\|\mathbf{y}_n - \hat{\mathbf{y}}_n\|_2}{\|\mathbf{y}_n\|_2}$, where $\mathbf{y}_n$ denotes the n th original frequency data under high SNR (≈ 40 dB) and without any interference, $\hat{\mathbf{y}}_n$ represents the corresponding recovery signal, and $N^* = 5200$ denotes the number of test samples. In order to show the small differences between RMSEs of the different methods more clearly, we use the logarithm coordinate for the vertical axes in Fig. 6.

As shown in Fig. 6(a), for the case without any interference, our CS inversions with $K = 2 \times L$ and $K = 4 \times L$ outperform other methods under each SNR case; the two versions of the OMP-based denoising method (Li et al., 2014) have the similar reconstruction performances under each SNR case; when $\text{SNR} \leq 15$ dB our statistical CS methods are obviously superior to the OMP-based denoising methods (Li et al., 2014), since our methods utilize the statistical information learned in the training stage beyond just sparseness, while the OMP only exploits the later. Since the denoising method in the noise-robust MTL-FA models (Du et al., 2012) also utilizes the statistical information, the reconstruction performance of our methods and the noise-robust MTL-FA models (Du

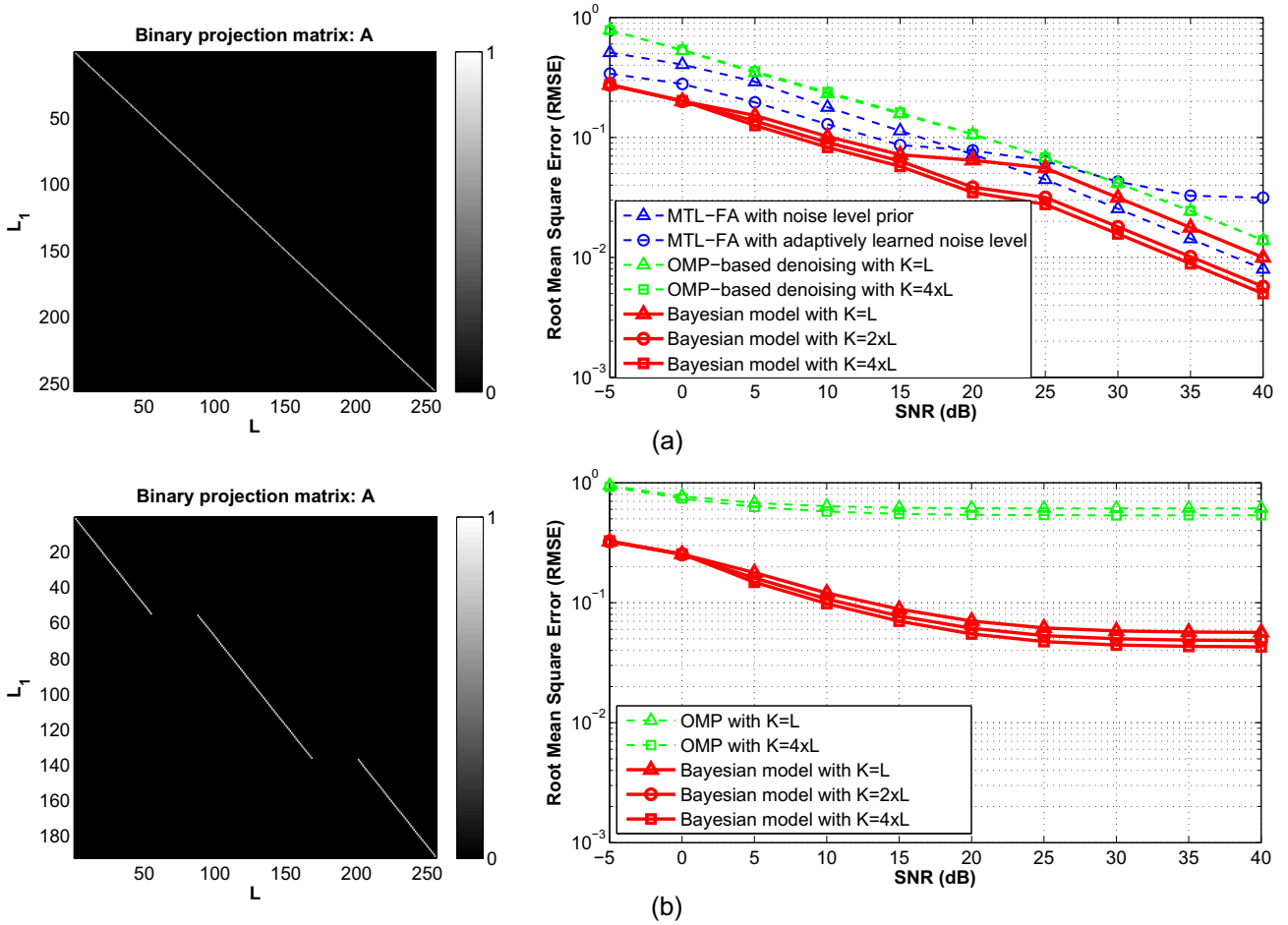


Fig. 6. Variation of the average reconstruction errors of the test data from the three airplane targets with SNR. (a) There is no interference, then the projection matrix $\mathbf{A}$ shown in the left subfigure is an identity matrix. Here we compare the denoising performances of the noise-robust MTL-FA model with the test noise level prior (Du et al., 2012); the noise-robust MTL-FA model with the adaptively learned test noise level (Du et al., 2012); the orthogonal matching pursuit (OMP) (Tropp & Gilbert, 2007) based denoising methods (Li et al., 2014) with $K = L$ and $K = 4 \times L$; our statistical CS inversions based on Bayesian models with $K = L$, $K = 2 \times L$ and $K = 4 \times L$. (b) The test samples are interfered via the projection matrix $\mathbf{A}$ shown in the left subfigure, where $\{\bar{y}^*(l)\}_{l=56}^{87}$ and $\{\bar{y}^*(l)\}_{l=169}^{200}$ are impacted by some interference signals. Here we compare the recovery performances of the OMP methods with $K = L$ and $K = 4 \times L$ and our statistical CS inversions based on Bayesian models with $K = L$, $K = 2 \times L$ and $K = 4 \times L$. In order to show the small differences between the reconstruction errors of the different methods more clearly, we use the logarithm coordinate for the vertical axes in the two right subfigures.

et al., 2012) mainly depends on the statistical learning. According to the derivation in Section 2.2, in our statistical model $\Lambda^{(d^*, m^*)} = \Phi \left(\frac{1}{N} \sum_{n=1}^N \mathbb{E}[\hat{\mathbf{w}}_n^{(d^*, m^*)}] \mathbb{E}[\hat{\mathbf{w}}_n^{(d^*, m^*)}]^H \right) \Phi^H + (\mathbb{E}[\mathbf{z}])^{-1} \mathbf{I}_L + \mathbb{E}[\hat{\mathbf{w}}_n^{(d^*, m^*)}]$ and $\mathbb{E}[\mathbf{z}]$ mainly depend on the distribution of dominant scattering centers in the target, i.e., $q(\mathbf{z}_n^{(d^*, m^*)})$. For a simple target, $q(\mathbf{z}_n^{(d^*, m^*)})$ may be easily learned with the orthogonal basis matrix Φ with $K = L$; while for a complicated target, e.g., an airplane, the super-resolution representation with $K > L$ may be needed to accurately learn $q(\mathbf{z}_n^{(d^*, m^*)})$. In the noise-robust MTL-FA models (Du et al., 2012), the statistical learning errors mainly come from the linear signal model assumption in the real-valued domain. As shown in Fig. 6(a), when $\text{SNR} > 20$ dB, the performance of our CS inversion with $K = L$ is a little worse than that of the noise-robust MTL-FA with the accurate noise level prior, since the statistical learning of our Bayesian method with $K = L$ is not so accurate and the linear signal model in the real-valued domain approximately holds under the high SNR condition.

Compared between Fig. 6(a) and (b), the reconstruction performance for the noised and interfered data is worse than that for the only noised data. Nevertheless, our CS methods are obviously superior to the OMP methods. If we assume that the RMSE threshold is 0.1, our methods can yield the better reconstruction

performance under the test condition of $\text{SNR} \geq 10$ dB for $K = 4 \times L$ and $\text{SNR} \geq 15$ dB for $K = 2 \times L$ and $K = L$; while the RMSEs of the OMP methods with $K = L$ and $K = 4 \times L$ are larger than 0.5 even under $\text{SNR} = 40$ dB for the narrowband interference case in the experiment. Here we should emphasize that the statistical information learned in the training stage beyond just sparseness guarantee the better reconstruction performance of our statistical CS method under low SNR or with fixed projection matrix corresponding to narrowband interference.

Fig. 7 gives the reconstruction examples of an HRR test sample from Cessna Citation S/II data for the narrowband interference case shown in Figs. 5(b) and 6(b), i.e., $\{\bar{y}^*(l)\}_{l=56}^{87}$ and $\{\bar{y}^*(l)\}_{l=169}^{200}$ are impacted by some interference signals. As shown in Fig. 7, when $\text{SNR} = 40$ dB, compared with the IFFT HRRP from the incomplete complex frequency response with $\{\bar{y}^*(l)\}_{l=56}^{87} = 0$ and $\{\bar{y}^*(l)\}_{l=169}^{200} = 0$, the reconstructed HRRP via the statistical CS inversion based on our Bayesian model with $K = 4 \times L$ can correctly recover some detailed structures in the reference HRRP under $\text{SNR} = 40$ dB and without any interference; when $\text{SNR} = 10$ dB or $\text{SNR} = 0$ dB, our statistical CS inversion results from the noised and incomplete data are also encouraged. The reconstructed HRR complex data can be further utilized to other applications.

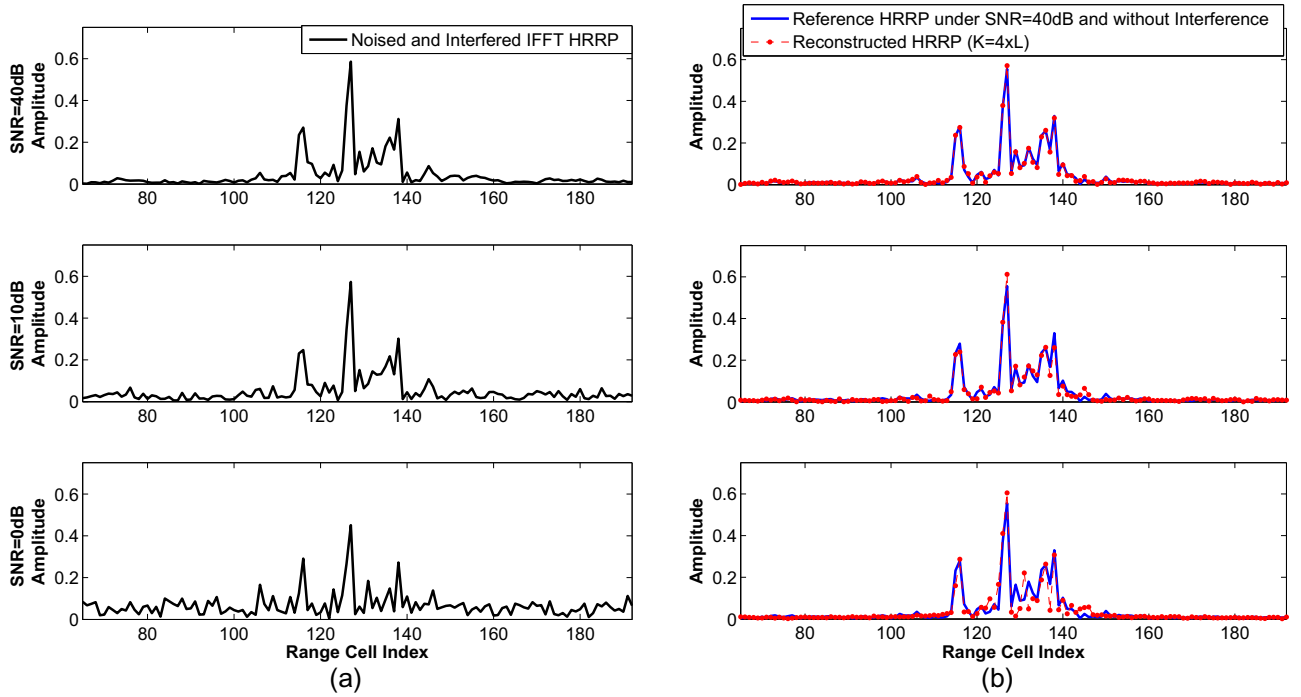


Fig. 7. Reconstruction examples of an HRR test sample from Cessna Citation S/II data for the narrowband interference case shown in Figs. 5(b) and 6(b), i.e., $\{\bar{y}^*(l)\}_{l=56}^{87}$ and $\{\bar{y}^*(l)\}_{l=169}^{200}$ are impacted by some interference signals. (a) The IFFT HRRPs from the noised and incomplete complex frequency responses with $\{\bar{y}^*(l)\}_{l=56}^{87} = 0$ and $\{\bar{y}^*(l)\}_{l=169}^{200} = 0$. (b) The reference HRRP under SNR = 40 dB and without any interference, and the reconstructed HRRPs via the statistical CS inversion based on our Bayesian model with $K = 4 \times L$. The different rows show the examples for the different SNR circumstances 40 dB, 10 dB and 0 dB.

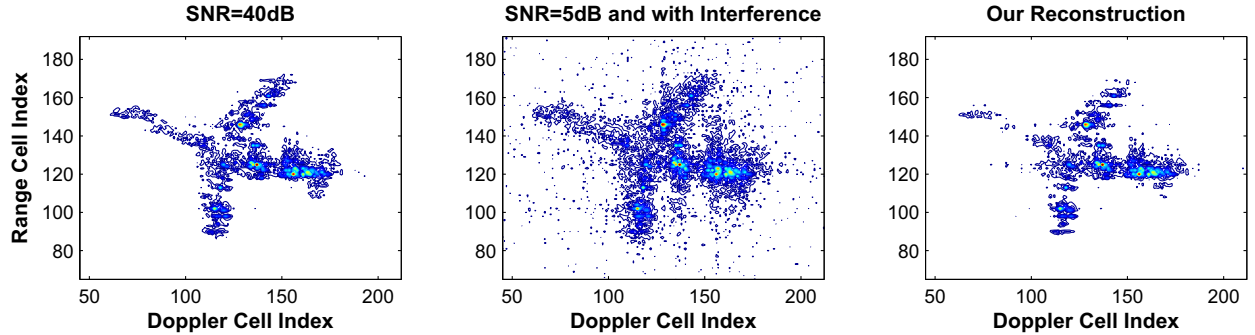


Fig. 8. ISAR imaging examples of an HRR frame from Yak-42 data. Left: The ISAR image generated via the HRR data under SNR = 40 dB and without any interference; Middle: The ISAR image generated via the HRR data under SNR = 5 dB and with the complex frequency responses $\{\bar{y}^*(l)\}_{l=56}^{87} = 0$ and $\{\bar{y}^*(l)\}_{l=169}^{200} = 0$, since $\{\bar{y}^*(l)\}_{l=56}^{87}$ and $\{\bar{y}^*(l)\}_{l=169}^{200}$ are impacted by some interference signals; Right: The ISAR image reconstructed from the HRR data in the middle subfigure via our statistical CS method ($K = 4 \times L$).

Fig. 8 gives the ISAR imaging examples of an HRR frame from Yak-42 data. Here our reconstruction example is performed in the 1-D HRR data domain, and the ISAR image is got via IFFT across the sequence of the reconstructed HRR signals. As shown in Fig. 8, our statistical CS inversion can obviously improve the ISAR imaging quality under low SNR and with narrowband interference. Therefore, although our Bayesian model is proposed for the 1-D HRR complex signal, the corresponding statistical CS method can also be applied to the 2-D SAR/ISAR imaging and recognition under low SNR or with narrowband interference.

4.6. Computation burden

All experiments have been performed in non-optimized software written in Matlab, on a Pentium PC with 1.50 GHz CPU and 2G RAM. For our measured HRR data, in the training stage, our Bayesian models with $K = L$, $K = 2 \times L$ and $K = 4 \times L$ respectively

required about 8 h (with about 60 VB iterations to get convergence), 20 h (with about 80 VB iterations to get convergence) and 50 h (with about 120 VB iterations to get convergence), all of which process the 35,840 training samples from all targets simultaneously. As shown in the above experiments, the recognition and reconstruction performance of our Bayesian model with $K = 4 \times L$ is best, but its computation burden is much huge. Therefore, in the real application, we need to strike an appropriate balance between time cost and recognition performance.

Although the above computation is pretty expensive, we know that the computation cost in the training stage can be ignored for an off-line learning (or training) system, and it is more important to evaluate the computation cost in the classification stage. If we calculate $\{\Lambda^{(d,m)}\}_{d=1,m=1}^{D,M_d}$ according to (15) in the training stage, the recognition and reconstruction procedures based on (17) and (22) are independent of the size of Φ . Therefore, the computation costs of our Bayesian models with different sizes of Φ should be

similar. Averagely, by our experiments, in the classification phase, our Bayesian models required about 2.2850 s to match a test sample with all frame models and reconstruct the denoised or complete test sample.

5. Conclusion

In last year's, various RATR studies based on HRR target signals were performed (Avci, 2013; Bharadwaj et al., 2001; Chen et al., 2008, 2009; Du et al., 2005, 2006, 2008b, 2011, 2012; Fasoula & van Genderen, 2011; Ferrara et al., 2009; Jacobs, 1999; Nelson et al., 2002; Shang et al., 2012; Shi et al., 2011; Webb, 2000; Williams et al., 2000; Zhang et al., 2012). Although scatterer distribution is an important information for RATR, these methods did not directly use such information. In this paper, we develop a hierarchical Bayesian model to infer the scattering coefficients and locations of the dominant scatterers with the automatic model selection, where a sparseness-promoting prior on the complex scattering coefficients and a Markov dependency for the location of the dominant scattering center between consecutive HRR signals are imposed. The results in Section 4.3 show that our signal model can obtain accurate parameter estimations, especially with the superresolution dictionary matrix Φ .

Based on such a signal model, a statistical recognition method for HRR complex data, which is robust to the condition of low SNR or narrowband interference, is proposed in this paper. There are two advantages of HRR complex data based RATR over HRR real data based RATR. Firstly, the HRR complex data can reserve some phase information, which should be useful in target discrimination. Secondly, the accurate modification of the statistical model can be easily derived in the complex-valued domain due to the linear signal model. Nevertheless, there has been little work done in the field of statistical recognition using HRR complex data. The results in Section 4.4 show that our statistical recognition method can obtain better performance than the traditional real-valued statistical recognition methods (Du et al., 2012; Jacobs, 1999; Li et al., 2014), especially under the relatively low SNR condition and with some narrowband interference. Although the recent work in Du et al. (2012) also proposes a noise-robust statistical recognition method, it is developed in the real-valued feature domain and cannot deal with the interference problem.

A statistical CS inversion is derived for reconstruction of the test sample under low SNR and with narrowband interference in this paper. The results in Section 4.5 show that our CS method is superior to the traditional CS methods (Du et al., 2012; Li et al., 2014) in the reconstruction performance due to utilizing the statistical information under high SNR and without any interference (beyond just sparseness).

In summary, this paper proposes a novel recognition and reconstruction scheme robust to noise and narrowband interference for the expert and intelligent systems community, which can make full use of the statistical structure and information in the complex signal domain. The experiments based on measured data show its good performance. In the real application, the proposed method is very effective for the online RATR due to its rapid response in the classification stage, as discussed in Section 4.6.

Nevertheless, the off-line learning (or training) system of our method requires the pretty expensive computation cost, especially in the case with the superresolution Φ . Therefore, in the real application, we need to strike an appropriate balance between time cost and recognition performance.

There also are some future work we can study on. An adaptive dictionary can be substituted for the fixed FFT dictionary matrix Φ in our Bayesian model, which may automatically learn the proper resolution for the given data. Current Bayesian model and statistical CS method can be extended for the 2-D SAR/ISAR images.

In this paper, we consider the CS reconstruction problem with a fixed projection matrix $\mathbf{A}$. In the expert and intelligent Systems community, the linear signal model $\bar{\mathbf{y}} = \mathbf{A}\mathbf{y} + \mathbf{v}$ can also be used as a feature extraction procedure. How to design an optimal projection matrix $\mathbf{A}$ to yield the most discriminative feature space is our future research direction. Furthermore, we can put some constraint on $\mathbf{A}$, e.g., $\mathbf{A} \in \{0, 1\}^{L_1 \times L}$ being a binary projection matrix with $L_1 < L$ and $\mathbf{A}\mathbf{A}^T = \mathbf{I}_{L_1}$, same as a projection matrix for the narrowband interference case. If the optimal $\mathbf{A}$ can be derived with such a constraint, then just a part of frequency components are useful for our recognition task and some resource can be saved in the real radar system.

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